Data Request: STADIOEquities Account Activation Project

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Module: CAP182 Capstone Project

Requested from: STADIOEquities data owner

Purpose: Build a binary-classification model that predicts, within 24 hours of registration, whether a newly registered account will make a successful first deposit within 30 calendar days.

1. Population, timing and delivery requirements

- **Population:** All new retail accounts registered during the most recent 24 complete months. Includes closed, rejected, incomplete, unfunded and funded accounts. Final registration cohort should end at least 30 days prior to the extract date so that every account has a complete outcome window.
- **Prediction time:** Registration time plus 24 hours. Predictor features are limited to those events which take place prior to this cut-off point.
- **Outcome:** `activated_30d = true` if the account has a successful, settled deposit greater than zero cents between registration and 30 calendar days post-registration; else `false`.
- **Time standard:** Provide timestamps in ISO 8601 format `yyyy-MM-ddTHH:mm:ss+02:00` with Africa/Johannesburg timezone, or else the source timezone and UTC offset.
- **Format:** Prefer parquet file format; UTF-8 csv is acceptable. Deliver one file per table, a data dictionary, extract date, name of source system, row count and control totals for registered and activated accounts.
- **Keys:** Substitute production account identifier with a stable pseudonymous `account_id` consistently used across all files. Primary keys should be unique; foreign keys need to resolve to the account table, with mismatch counts provided.
- **Missing data:** Preserves genuine nulls. Do not substitute unknown values with zero, blank text or invented categories. State the null code and reason where available.
- **Privacy and security:** Avoid names, identity numbers, exact birth dates, email addresses, phone numbers, physical addresses, bank-account identifiers, credentials, identity-document images or any unredacted free text. Ensure delivery encrypted in transit and at rest, with access limited to the project team only and retention and deletion agreement followed.
- **Leakage control:** Outcomes, outcome-building and post-cut-off fields will be requested for labelling and validation but not used as predictors. Permitted use of these is labelled below.

2. Table 1: Account and funding

Grain: One row per registered account, including incomplete and never-funded accounts.

Column name	Data type	Example	Additional information
Account ID	String, identifier	ACC-0093214	Required; unique pseudonymous key. Must match the account key used in every other table.

Column name	Data type	Example	Additional information
Registration time	String, datetime	2025-03-14T09:21:07+02:00	Required ISO 8601 timestamp for account creation; this starts the 24-hour feature and 30-day outcome windows.
Registration status	String, categorical	complete	Required; valid values: `incomplete`, `complete`, `rejected`, `cancelled`, `closed`. Provide mapping for any source codes.
Account tier	String, categorical	standard	Tier selected at registration; valid values and business definition must be provided. Null only if no tier was selected.
KYC status 24h	String, categorical	pending	Latest KYC status at the 24-hour prediction cut-off; valid values: `not_started`, `in_progress`, `pending_review`, `passed`, `failed`.
KYC completion time	String, datetime, nullable	2025-03-14T10:02:18+02:00	Timestamp when KYC first passed. Null if it had not passed. May be used as a predictor only if at or before the 24-hour cut-off.
Funding event ID	String, identifier, nullable	DEP-800421	Identifier of the first deposit attempt. Null if no attempt was made. Link to the funding event record if a separate transaction file is supplied.
First deposit time	String, datetime, nullable	2025-03-16T13:41:20+02:00	Timestamp of the first successful settled deposit. Outcome-building field; never use as a predictor. Null if no successful deposit has taken place.
First deposit amount	Numeric, integer, nullable	25000	First successful deposit in South African cents; must be greater than 0. Outcome-validation field; never use as a predictor.
First deposit status	String, categorical	settled	Valid values: `none`, `initiated`, `pending`, `failed`, `reversed`, `settled`. State how reversals are handled in the target.
Activated 30d	Boolean	true	Derived target: `true` only if a successful settled deposit greater than 0 took place within 30 days of registration; else `false`.
Activation deadline	String, datetime	2025-04-13T09:21:07+02:00	Registration time plus 30 calendar days; this is requested in order to enable audit of the outcome-window logic.
Current balance	Numeric, integer	185450	Account cash plus holdings value in South African cents at extract time. For validation only; never use as an activation predictor.
Closure time	String, datetime, nullable	2025-06-02T11:08:00+02:00	Time account was closed or cancelled. Null if account is still open. Closure reason should be supplied as a coded category where possible.
Extract time	String, datetime	2026-08-31T08:00:00+02:00	Required snapshot timestamp shared by all files or else clearly documented per source.

3. Table 2: App and web behaviour

Grain: One row per app or web session. Sessions from registration up to 30 days after; only sessions started within the first 24 hours can become predictors.

Column name	Data type	Example	Additional information
Session ID	String, identifier	SES-733190	Required; unique per session.
Account ID	String, identifier	ACC-0093214	Required foreign key to Table 1. Unlinked sessions must be reported.
Session start	String, datetime	2025-03-14T09:21:09+02:00	Required ISO 8601 session-start timestamp.
Session end	String, datetime, nullable	2025-03-14T09:29:47+02:00	Null only if session did not end normally; please explain timeout handling.
Platform	String, categorical	android_app	Valid values: `android_app`, `ios_app`, `mobile_web`, `desktop_web`, `other`.
OS family	String, categorical	Android	Operating system family; use controlled list and group rare values under `other`. No device advertising identifiers allowed.
App version	String, categorical, nullable	8.4.1	Version of mobile app or web release; null if not recorded in the source.
Entry source	String, categorical	campaign_link	Source of the session start. Valid values should include `direct`, `campaign_link`, `email`, `push`, `referral`, `organic_search`, `other`.
Onboarding step	String, categorical	bank_link	Furthest onboarding step completed in the session. Provide ordered onboarding step dictionary, keeping `not_started` and `completed` values.
Onboarding complete	Boolean	false	`true` only if all mandatory onboarding steps have been completed by session end.
Screens viewed	Numeric, integer	9	Count of screen-view events in the session; minimum 0. Explain handling of bots and duplicate events.
Feature count	Numeric, integer	4	Number of distinct product features used in the session; minimum 0. Provide feature names dictionary separately where available.
Duration seconds	Numeric, integer, nullable	518	Session duration in seconds; minimum 0. Null if no reliable session end could be inferred.
Abandoned	Boolean	true	`true` when the user leaves an incomplete mandatory onboarding flow and does not resume within the agreed timeout. State the timeout rule.

Column name	Data type	Example	Additional information
Error count	Numeric, integer	2	Count of client or server error events in the session; range 0 or more. Exclude harmless diagnostic messages.

4. Table 3: Client demographics and stated needs

Grain: One row per account using values known at registration. Values must be coarsened enough to protect identity.

Column name	Data type	Example	Additional information
Account ID	String, identifier	ACC-0093214	Required foreign key to Table 1.
Age band	String, categorical	25-34	Age at registration, not date of birth. Valid values: `18-24`, `25-34`, `35-44`, `45-54`, `55-64`, `65+`, `unknown`.
Province	String, categorical	GP	Valid values: `EC`, `FS`, `GP`, `KZN`, `LP`, `MP`, `NC`, `NW`, `WC`, `outside_SA`, `unknown`.
Area type	String, categorical	urban	Valid values: `urban`, `peri_urban`, `rural`, `unknown`; derive before delivery and do not supply a street address.
Preferred language	String, categorical	isiZulu	Controlled list of supported languages plus `other` and `unknown`.
Employment status	String, categorical	employed	Valid values: `employed`, `self_employed`, `student`, `unemployed`, `retired`, `other`, `prefer_not_to_say`, `unknown`.
Income band	String, categorical	R10001-R25000	Monthly gross-income band captured at sign-up. Supply ordered band definitions; do not provide exact income.
Investing experience	String, categorical	beginner	Valid values: `none`, `beginner`, `intermediate`, `experienced`, `unknown`.
Stated goal	String, categorical	long_term_growth	Valid values should include `emergency_saving`, `education`, `home`, `retirement`, `long_term_growth`, `short_term_trading`, `other`, `unknown`.
Risk appetite	String, categorical	moderate	Client-stated or questionnaire result at registration; valid values: `low`, `moderate`, `high`, `not_completed`, `unknown`.
Tax residency	String, categorical	South_Africa_only	Coarse valid values: `South_Africa_only`, `multiple`, `other`, `unknown`; no tax numbers.

5. Table 4: Marketing and communication events

Grain: One row per attributable acquisition touchpoint or onboarding communication. Include 30 days before registration through 30 days after; only events at or before the 24-hour cut-off will be predictors.

Column name	Data type	Example	Additional information
Event ID	String, identifier	MKT-400821	Required; unique per touchpoint or message event.
Account ID	String, identifier	ACC-0093214	Foreign key to Table 1; may be null for pre-registration anonymous touches, but provide a stable pseudonymous journey ID if linkable later.
Event time	String, datetime	2025-03-14T08:54:12+02:00	Required ISO 8601 timestamp.
Channel	String, categorical	paid_social	Valid values should include `paid_search`, `paid_social`, `organic`, `referral`, `affiliate`, `email`, `sms`, `push`, `in_app`, `direct`, `other`.
Campaign ID	String, identifier, nullable	CMP-2025-031	Stable campaign key with a separate campaign-name dictionary; null for direct or organic events.
Creative ID	String, identifier, nullable	CR-184	Pseudonymous creative or message-variant key; null where not applicable.
Event type	String, categorical	click	Valid values: `impression`, `click`, `registration_attributed`, `message_sent`, `delivered`, `opened`, `clicked`, `bounced`, `unsubscribed`.
Attribution type	String, categorical, nullable	last_touch	Valid values: `first_touch`, `last_touch`, `multi_touch`, `unattributed`; state attribution window and rule.
Cost cents	Numeric, integer, nullable	18000	Attributed acquisition or contact cost in South African cents; 0 or more. State whether account-level cost is actual or allocated.
Incentive code	String, categorical, nullable	WELCOME50	Controlled promotion identifier; null when no incentive applied. Do not include free text.
Consent status	String, categorical	opted_in	Channel-specific status at event time; valid values: `opted_in`, `opted_out`, `not_asked`, `not_applicable`.
Delivery status	String, categorical, nullable	delivered	Valid values: `queued`, `sent`, `delivered`, `failed`, `bounced`, `not_applicable`.

6. Table 5: KYC and compliance process events

Grain: One row per KYC or account-verification event from registration through 30 days afterwards. This request concerns onboarding friction, not transaction-monitoring investigations.

Column name	Data type	Example	Additional information
Compliance event ID	String, identifier	KYC-900118	Required; unique per verification event.
Account ID	String, identifier	ACC-0093214	Required foreign key to Table 1.
Event time	String, datetime	2025-03-14T09:37:52+02:00	Required ISO 8601 timestamp.
Process step	String, categorical	identity_check	Valid values should include `identity_check`, `address_check`, `bank_check`, `selfie_check`, `manual_review`, `final_decision`.
Event status	String, categorical	retry_required	Valid values: `started`, `submitted`, `pending`, `passed`, `failed`, `retry_required`, `expired`.
Reason code	String, categorical, nullable	image_unreadable	Controlled failure or delay code. Null for successful events. Do not provide investigator notes or document contents.
Document type	String, categorical, nullable	SA_ID_card	Coarse document category only; do not supply document numbers, images or extracted personal text.
Attempt number	Numeric, integer	2	Sequential attempt count for the account and process step; minimum 1.
Manual review	Boolean	false	`true` if a human review was required for this event.
Queue seconds	Numeric, integer, nullable	1460	Time from submission to a decision in seconds; 0 or more. Null if the event remains open.
Final result	String, categorical, nullable	passed	Valid values: `passed`, `failed`, `withdrawn`, `pending`; populated only for a final-decision event.

7. Table 6: Support and feedback interactions

Grain: One row per support ticket or structured feedback event from registration through 30 days afterwards. Only events by the 24-hour cut-off will be predictors.

Column name	Data type	Example	Additional information
Interaction ID	String, identifier	TKT-512903	Required; unique per ticket or feedback event.
Account ID	String, identifier	ACC-0093214	Required foreign key to Table 1 where the interaction is from a registered user.
Opened time	String, datetime	2025-03-14T10:14:21+02:00	Required ISO 8601 creation timestamp.

Column name	Data type	Example	Additional information
Closed time	String, datetime, nullable	2025-03-14T13:51:09+02:00	Null if unresolved at extract time.
Channel	String, categorical	in_app_chat	Valid values: `in_app_chat`, `email`, `phone`, `web_form`, `app_review`, `other`.
Reason category	String, categorical	deposit_help	Controlled categories should include `registration`, `KYC`, `deposit_help`, `fees`, `product_question`, `technical_error`, `withdrawal`, `other`.
Ticket status	String, categorical	resolved	Valid values: `open`, `pending_client`, `pending_internal`, `resolved`, `closed`.
First response seconds	Numeric, integer, nullable	420	Seconds from opening to first staff response; 0 or more. Null when there has been no response.
Resolution seconds	Numeric, integer, nullable	13008	Seconds from opening to resolution; 0 or more. Null when unresolved.
Contact count	Numeric, integer	3	Number of messages or calls in the interaction; minimum 1.
CSAT score	Numeric, integer, nullable	4	Customer satisfaction score from 1 to 5. Null when no survey response was received.
Sentiment class	String, categorical, nullable	negative	If already approved and derived by the client: `positive`, `neutral`, `negative`, `unknown`. Do not supply raw free text by default.

8. Table 7: Product and subscription selections

Grain: One row per account-product relationship or subscription-status change. Include choices from registration through 30 days afterwards; use only choices made by the 24-hour cut-off as predictors.

Column name	Data type	Example	Additional information
Relationship ID	String, identifier	PRD-771240	Required; unique per account-product event or snapshot row.
Account ID	String, identifier	ACC-0093214	Required foreign key to Table 1.
Event time	String, datetime	2025-03-14T09:45:33+02:00	Required ISO 8601 timestamp for selection or status change.
Product ID	String, identifier	TAXFREE-01	Stable product key; provide a separate product dictionary.

Column name	Data type	Example	Additional information
Product category	String, categorical	tax_free_account	Valid values should include `standard_account`, `tax_free_account`, `offshore`, `bundle`, `premium_subscription`, `other`.
Event type	String, categorical	selected	Valid values: `viewed`, `selected`, `started`, `activated`, `cancelled`, `expired`.
Subscription tier	String, categorical, nullable	premium	Valid values: `none`, `standard`, `premium`, plus any current client-defined tiers.
Goal amount cents	Numeric, integer, nullable	1000000	Optional client-stated goal in South African cents; 0 or more. Null when not requested or not supplied.
Recurring deposit	Boolean	true	Whether the client set up a recurring deposit instruction by event time.
Recurring amount	Numeric, integer, nullable	50000	Planned recurring amount in South African cents; greater than 0 when recurring deposit is true, otherwise null.
Cancellation reason	String, categorical, nullable	price	Controlled reason code; null unless the event type is `cancelled`. No free text.

9. Table 8: Funding attempts and payment outcomes

Grain: One row per deposit attempt from registration through 30 days afterwards. This is the authoritative target-building table.

Column name	Data type	Example	Additional information
Deposit ID	String, identifier	DEP-800421	Required; unique per deposit attempt.
Account ID	String, identifier	ACC-0093214	Required foreign key to Table 1.
Initiated time	String, datetime	2025-03-16T13:39:02+02:00	Required ISO 8601 timestamp when the deposit was initiated.
Settled time	String, datetime, nullable	2025-03-16T13:41:20+02:00	Timestamp when funds became available. Null unless status is `settled`. Used to construct the target.
Amount cents	Numeric, integer	25000	Deposit amount in South African cents; greater than 0.
Payment method	String, categorical	instant_EFT	Valid values should include EFT, `instant_EFT`, `debit_order`, `card`, `bank_transfer`, `other`. Do not provide bank details or payment tokens.

Column name	Data type	Example	Additional information
Deposit status	String, categorical	settled	Valid values: `initiated`, `pending`, `failed`, `reversed`, `settled`. Supply status-change logic.
Failure code	String, categorical, nullable	bank_declined	Controlled failure code; null for settled deposits. No free-text payment messages.
Reversed flag	Boolean	false	`true` when a previously settled deposit was reversed within the 30-day outcome window. State whether such cases are excluded or labelled inactive.
Attempt number	Numeric, integer	1	Sequential deposit-attempt count per account; minimum 1.

10. Quality checks requested with delivery

Please provide the following controls so the extract can be reconciled before modelling:

1. Row count, distinct primary-key count and duplicate-key count for every table.
2. Minimum and maximum event timestamp for every event table.
3. Null percentage for every requested column, with source-system missing-value codes documented.
4. Registered-account count by calendar month and registration status.
5. Count and percentage of accounts with a settled positive first deposit within 30 days, by registration month.
6. Count of orphan foreign keys in each child table and the reason for any expected unmatched records.
7. A versioned data dictionary containing source field, transformation, valid values, units, owner and refresh frequency.
8. A change log for major app, onboarding, KYC, campaign or payment-process changes during the requested period.

11. Acceptance criteria

The delivery will be accepted for initial analysis when all files can be opened, primary keys are unique at the stated grain, account IDs reconcile across sources, timestamps and monetary units are unambiguous, every included registration has a mature 30-day outcome window, target control totals match the client’s reported totals within an explained tolerance, and no prohibited direct identifiers are present.